Research Statement

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Information systems that complement social infrastructure

My research examines the mechanisms through which information systems complement existing human services and improve social outcomes. I focus on mobility and healthcare because both make a common problem visible: a technically capable system can exist without being accessible, trusted, or useful to the people who need it. I therefore connect artifact design with behavioral explanation and empirical evaluation. I want to understand not simply whether a technology works, but for whom it creates value, through which mechanisms, and under what geographic and institutional conditions.

As a Ph.D. student in Management Information Systems within the Business and Technology Management program at KAIST, advised by Professor Chul Ho Lee, I develop this agenda across technical, behavioral, and societal levels. My projects move between building systems, studying how people interpret and use them, and examining their consequences in real settings. Mobility provides the spatial foundation of this agenda; healthcare adds questions of vulnerability, trust, and access.

Mobility as an information systems artifact

My technical mobility research combines OpenStreetMap road data with satellite-image analysis to support road detection and recommended-speed calculation. I have taken a leading role in algorithm development and related patent work. This includes road segmentation and road-graph construction, geometric estimation of curvature, and the combination of physics-based speed constraints with XGBoost recommendations. Two Korean patent applications concern road determination and recommended-speed determination. This work taught me to treat map quality, model uncertainty, and operational constraints as integral parts of an artifact's practical value.

At the behavioral level, my manuscript "Understanding Personal Mobility Device Adoption Through the S–O–R Framework: A Multi-Group Analysis of Users and Non-users" examines 444 respondents in South Korea, evenly divided between users and non-users. Using PLS-SEM and multi-group analysis, I study how perceived risk and trust connect mobility-related beliefs to usage intention. In the current analysis, trust and physical risk are important in both groups, while the ease-of-use-to-risk relationship differs across groups. The manuscript is under review. These findings direct my attention to the distinction between anticipated benefits and experience-based evaluations; because the survey is cross-sectional, I interpret the pathways as behavioral relationships rather than established causal effects.

At the societal level, I secured and led a local-government-supported project for Yuseong District on public bike-sharing, transit integration, local festivals, and consumer activity. The project combined Tashu bike-sharing and transit information with Shinhan Card spending data and KT population-movement data. It examines first- and last-mile connections, flows between places, and the conditions under which mobility translates into local economic activity. The work received an Excellent Paper Award from the Korea Society of IT Services and is under revision at a Korean journal. It strengthened my interest in community well-being while also showing why mobility, spending, and welfare must remain distinct empirical outcomes.

Healthcare and the conditions for engagement

My healthcare work asks how information systems can reduce barriers to engagement in psychologically sensitive settings. In collaboration with Jeonbuk National University Hospital, I have developed an LLM- and retrieval-augmented-generation chatbot that is offered free of charge to patients with schizophrenia. A motivating observation is that some patients may feel less exposed to interpersonal judgment when interacting with a nonhuman system. I investigate when this affordance supports disclosure and engagement, and when design choices may instead produce distrust or perceived judgment. This is a question about the conditions of support, not a presumption that a chatbot replaces clinical care or has already demonstrated a clinical treatment effect.

My "Thought-Bubbles in Therapy" research studies how visible reflective or explanatory messages in conversational interfaces influence perceived judgmentalness, authenticity, trust, and disclosure. The displayed messages are an interface design intervention; they should not be equated with direct access to a model's internal reasoning. The broader program includes online experimental work and patient-facing development, with research dissemination through CIST, ICIS, and SCECR. Across these settings, I am interested in capabilities that make AI useful alongside human professionals, rather than assuming that greater human likeness is always desirable.

Connecting GIS and healthcare through drone delivery

My next project connects these streams by asking whether drone-enabled medication delivery reduces geographic barriers to care in the United States. A prescription has limited practical value if obtaining it requires a difficult journey, access to a vehicle, or time that a patient cannot readily spare. Autonomous delivery may shorten that journey, but nominal coverage does not necessarily create usable access. Eligibility rules, service interruptions, trust, digital ordering requirements, and the availability of participating pharmacies may determine who actually benefits.

I propose to begin with the measurement problem: establishing where and when prescription delivery was available. Historical Planet imagery could characterize known facilities and surrounding development and, where features are observable, independently corroborate physical changes. Documentary sources must establish actual service operation. I will distinguish facility construction, regulatory approval, service commencement, and closure. A targeted feasibility pilot will determine whether PlanetScope provides sufficient detail or whether a separate high-resolution archive sample is required.

The eventual empirical design will combine a validated service registry, OSM network-accessibility measures, demographic characteristics, and outcome data obtained through appropriate agreements. Where rollout and comparison conditions permit, cohort-specific difference-in-differences and event studies will examine changes in access and use. I will assess heterogeneity by baseline transport and pharmacy-access constraints. Clinical effects will be studied only with suitable longitudinal data; spending and mobility measures will not substitute for health outcomes. This approach treats satellites as a measurement resource within an information systems research design.

Implementation experience and research practice

Since February 2024, I have worked full-time as Chief Data Officer at TalentSeeker, a startup where I lead work on data ontologies, multi-agent systems, data-pipeline architecture, data engineering, and LLMOps. Government-supported recruitment R&D has given me experience translating research ideas into operating systems and evaluating data quality, retrieval, orchestration, and inference workflows. My additional project experience includes a 2026 international smart-farm research plan, with a role in data experimentation and bias analysis. These activities complement my academic work by making reproducibility, monitoring, and the limits of available data concrete.

I also teach data science as the lead course instructor at Baekseok University. My course covers Python and pandas, data preparation, visualization, basic statistics and prediction, responsible use of generative AI, and communication of analytical results. Teaching reinforces a principle that guides my research: analytical sophistication is useful only when the data, assumptions, and interpretation can be made clear.

Long term research direction

My long-term goal is to explain how digital artifacts become socially valuable through their interaction with people and institutions. In mobility, I ask how a new mode changes access to places and services. In healthcare, I ask how an interface or delivery system changes the willingness and ability to obtain care. In both cases, I seek mechanisms of complementarity, including the conditions under which those mechanisms fail or benefits remain unevenly distributed.

The Planet proposal is a concrete next step in this agenda. It brings together geospatial measurement, technical implementation, behavioral reasoning, and careful outcome evaluation. I hope to produce research that helps service designers and policymakers decide not only where to deploy a technology, but how to connect it to existing systems so that its value reaches the people it is intended to serve. Any Planet data access would be used exclusively for non-commercial academic research at KAIST, separate from TalentSeeker’s commercial operations.